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## Experimental study of the Timoshenko beam theory predictions

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## ABSTRACT

The theory of flexural vibrations proposed by Timoshenko almost 90 years ago has been the subject of several recent papers. In the Timoshenko beam theory a critical frequency  $f_c$  is expected and for frequencies  $f$  larger than  $f_c$ , some authors argue that a second spectrum exists. This is still the matter of much debate. In order to settle this question we have performed experimental measurements of flexural normal-mode frequencies and amplitudes which are much more precise than what has been obtained up to now. For some purposes cylindrical rods are more convenient, whereas in other cases rectangular beams are easier to deal with, so we analyze rods and beams of different shapes and sizes. We show that, below  $f_c$ , all observed resonant peaks can be associated with one of the compressional, torsional or Timoshenko resonances and that no resonant peaks are left undetected. The validity of Timoshenko beam theory below  $f_c$  and that only one spectrum appears in this regime is therefore proved. The comparison between theory and experiment for rectangular beams also shows that our experimental setup is appropriate, so we can use it confidently for  $f > f_c$ . The measurements we report here in this regime show the existence of a second spectrum, at least for free-free boundary conditions.

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## 1. Introduction

The study of flexural vibrations in beams and rods is an important problem in elasticity since these vibrating quasi-one-dimensional systems are ubiquitous [1,2]. Flexural vibrations were first described by Bernoulli and Euler in the eighteenth century. Their theory leads to a fourth-order differential equation for the transversal displacement. Unfortunately, a dispersion relation is obtained for which the wave velocity grows quadratically with frequency. Given this unphysical behavior, it is expected that the Bernoulli–Euler theory be valid only at low frequencies. In the nineteenth century, the problem was considered again by Rayleigh, who took into account rotational inertia. Almost 90 years ago, Timoshenko Beam Theory (TBT) was established [3]. This theory agrees with the Bernoulli–Euler results for the lower normal modes but it fits experimental data at higher frequencies as it is well known and we have proved experimentally for a rod with free-free boundary conditions [4].

At higher frequencies, the Timoshenko beam theory presents a critical frequency  $f_c$ , for which the equation is drastically altered [5,6]. For values of the frequency  $f < f_c$ , TBT seems to agree well with theoretical and numerical calculations [7–10] as well as with experimental measurements of the normal-mode spectrum [4,11,12].

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However, for  $f > f_c$  this is no longer the case, and what was called the second spectrum of TBT has been the subject of a long standing debate [11,13–23]. Up to now it is not clear that the TBT predictions, above the cut-off frequency, are valid. Experimental observations are needed to settle this point definitely. This is precisely what we intend to do in this paper. One should point out that experimental results are scarce. For example, Chan et al. [7] analyzed a beam with free–free boundary conditions, which are the easiest boundary conditions to be dealt with experimentally; however, only a few measured frequency points were obtained. The system studied in Ref. [11], unfortunately, is far from being quasi-one-dimensional. Our aim is to extend these measurements. Using the electromagnetic acoustic transducer (EMAT) that we have developed [24] we are able to measure normal-mode frequencies and amplitudes for vibrating rods and beams.

As we will show in Section 3, this EMAT is very selective and distinguishes among compressional, torsional and flexural vibrations. Notice that the transducer works through Lorentz force on the electrons of the aluminum but not through magnetic force since the aluminum is diamagnetic [24]. By comparing the experimental spectrum with the well known values for compressional and torsional waves, and with theoretical values obtained for TBT and free-end boundary conditions in Section 2, we are able to identify with a very small error the nature of all possible vibrations of a rod for  $f < f_c$ . This is most easily done for a cylindrical rod, since the flexural eigenmodes appear as doublets. As we can also measure the wave amplitudes, we are able to obtain the dispersion relation and compare it with the Timoshenko predictions. In both comparisons the experimental agreement with TBT is excellent for  $f < f_c$ , as shown in Section 4.

Once we are convinced that our experimental method renders very precise results, we turn to analyze the case  $f > f_c$  and discuss whether the second spectrum exists or not. In Section 5 we present the comparison between TBT, finite element method (FEM) results, and experiment and show that, indeed, the second spectrum exists, at least for the free-end situation.

## 2. Timoshenko beam theory, solutions and boundary conditions

Besides the transversal displacement  $\xi$  Timoshenko introduced an angular variable  $\Psi$ , as indicated in Fig. 1. The second-order differential equations satisfied by  $\Psi$  and  $\xi$ , governing the flexural behavior of a beam of uniform cross-section, are [26]:

$$\kappa GA \left( \frac{\partial \Psi}{\partial z} + \frac{\partial^2 \xi}{\partial z^2} \right) - \rho A \frac{\partial^2 \xi}{\partial t^2} = 0, \quad (1)$$

$$\kappa GA \left( \frac{\partial \xi}{\partial z} + \Psi \right) - EI \frac{\partial^2 \Psi}{\partial z^2} + \rho I \frac{\partial^2 \Psi}{\partial t^2} = 0. \quad (2)$$

Here  $E$  is Young's modulus,  $G$  the shear modulus,  $\rho$  the mass density,  $I$  the second moment of area, and  $A$  the cross-sectional area;  $\kappa$  is called the Timoshenko shear coefficient, and it is the only free parameter included in this theory. Eqs. (1) and (2) were obtained under the assumption that the surface  $\Pi_c$  in Fig. 1 can be approximated by a flat surface  $\Pi_p$ . The previous equations can be written as a fourth-order differential equation for the transversal displacement  $\xi$  [1],

$$\frac{EI}{\rho A} \frac{\partial^4 \xi}{\partial z^4} - \frac{I}{A} \left( 1 + \frac{E}{\kappa G} \right) \frac{\partial^4 \xi}{\partial z^2 \partial t^2} + \frac{\partial^2 \xi}{\partial t^2} + \frac{\rho I}{\kappa GA} \frac{\partial^4 \xi}{\partial t^4} = 0, \quad (3)$$

known as Timoshenko's equation. The angular variable  $\Psi$  satisfies the same equation [1]. One should notice that there exist other theories including more effects like the inertia correction factor of Ref. [25].

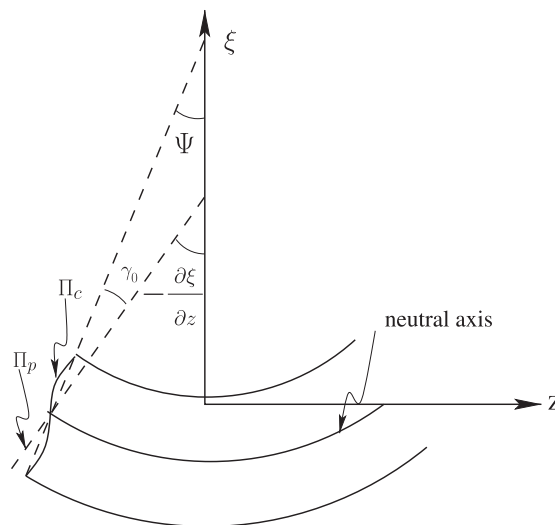


Fig. 1. Definition of the angular variable  $\Psi$  and transversal displacement  $\xi$ .

## 2.1. Solutions of Timoshenko's equation

The solution to Eq. (3) can be obtained separating the variables as follows. We insert the ansatz

$$\xi(z,t) = e^{i\omega t} \chi(z) \quad (4)$$

in Eq. (3), where  $i = \sqrt{-1}$  is the imaginary unit and the angular frequency  $\omega = 2\pi f$ . This yields

$$\frac{d^4 \chi}{dz^4} - \alpha \frac{d^2 \chi}{dz^2} + \frac{\beta}{4} \chi = 0, \quad (5)$$

where we have defined

$$\alpha = -\frac{\rho \omega^2}{M_r}, \quad (6)$$

$$\beta = \frac{4\rho^2 \omega^2}{\kappa GE} (\omega^2 - \omega_c^2), \quad (7)$$

with  $M_r$  the “reduced modulus”

$$\frac{1}{M_r} = \frac{1}{E} + \frac{1}{\kappa G} \quad (8)$$

and

$$\omega_c = 2\pi f_c = \sqrt{\frac{\kappa GA}{\rho I}}. \quad (9)$$

The characteristic equation associated to Eq. (5) is

$$k^4 - \alpha k^2 + \frac{\beta}{4} = 0, \quad (10)$$

with solutions

$$\begin{aligned} k_{\pm}^2(\omega) &= \frac{\alpha \pm \sqrt{\alpha^2 - \beta}}{2} \\ &= -\frac{2\pi^2 \rho f^2}{M_r} \pm 2\pi^2 \rho f \sqrt{\frac{f^2}{M_r^2} - \frac{4(f^2 - f_c^2)}{\kappa GE}} \\ &= -\frac{\rho \omega^2}{2M_r} \pm \frac{1}{2} \sqrt{\frac{\rho^2 \omega^4}{M_r^2} - \frac{4\rho^2 \omega^2}{\kappa GE} (\omega^2 - \omega_c^2)}, \end{aligned} \quad (11)$$

which are the dispersion relations. From this equation one can see that the solutions change its behavior depending on the sign of  $\beta$  which in turn will depend on the value of  $\omega$ . As we will show,  $\omega_c$  separates the solutions into two regimes:  $\omega < \omega_c$  and  $\omega > \omega_c$ . On the one hand, from Eq. (7) one sees that when  $\omega < \omega_c$ ,  $\beta$  is negative. Then  $\alpha + \sqrt{\alpha^2 - \beta} > 0$  so the two values of  $k_+$  are real. This yields exponential solutions of Eq. (5). One can also prove that both values of  $k_-$  are imaginary in this regime. On the other hand, when  $\omega > \omega_c$ , it is possible to show that both  $k_-$  and  $k_+$  are imaginary.

The general solution to Eq. (5) for all frequencies is finally given by

$$\chi(z) = a \exp(k_+ z) + b \exp(-k_+ z) + c \exp(ik_- z) + d \exp(-ik_- z), \quad (12)$$

where  $a$ ,  $b$ ,  $c$ , and  $d$  are constants to be determined by the boundary conditions.

## 2.2. Boundary conditions

In this subsection we study a rod of length  $L$  with free–free boundary conditions since these boundary conditions can be easily obtained in the laboratory [4,12]. The boundary conditions for a beam with free ends are given by the vanishing of moments and shear forces [5,26],

$$\left. \frac{\partial \Psi}{\partial z} \right|_{z=0,L} = 0, \quad (13)$$

$$\left. \left( \frac{\partial \xi}{\partial z} + \Psi \right) \right|_{z=0,L} = 0, \quad (14)$$

respectively. Since Eq. (3) is given only for one variable, say  $\xi$  or  $\Psi$ , these boundary conditions should be changed to the corresponding variable when working with that equation. To write the time independent version of Eqs. (1) and (2) we

need also a separation of variables for the angle  $\Psi$  as follows:

$$\Psi(z,t) = e^{i\omega t} \psi(z). \quad (15)$$

Inserting Eqs. (4) and (15) into Eqs. (13) and (14) one gets

$$\left. \frac{d\psi}{dz} \right|_{z=0,L} = 0, \quad (16)$$

$$\left( \frac{d\chi}{dz} + \psi \right) \Big|_{z=0,L} = 0. \quad (17)$$

Substituting Eqs. (4) and (15) into Eqs. (1) and (2) one gets

$$\left( \frac{d\psi}{dz} + \frac{d^2\chi}{dz^2} \right) + \frac{\rho\omega^2}{\kappa G} \chi = 0, \quad (18)$$

$$\kappa GA \left( \frac{d\chi}{dz} + \psi \right) - EI \frac{d^2\psi}{dz^2} - \rho I \omega^2 \psi = 0. \quad (19)$$

The first boundary condition can be obtained by evaluating Eq. (18) at  $z=0, L$  and using Eq. (16). One gets

$$\left. \frac{d^2\chi}{dz^2} \right|_{z=0,L} = - \frac{\rho\omega^2}{\kappa G} \chi \Big|_{z=0,L}. \quad (20)$$

From Eqs. (17) and (19) another boundary condition for  $\psi$  is obtained,

$$\left. \frac{d^2\psi}{dz^2} \right|_{z=0,L} = - \frac{\rho\omega^2}{E} \psi \Big|_{z=0,L}. \quad (21)$$

Other two boundary conditions can be obtained taking the derivative of Eq. (18) and using Eq. (21). This yields

$$\left( - \frac{\rho\omega^2}{E} \psi + \frac{d^3\chi}{dz^3} \right) \Big|_{z=0,L} = - \frac{\rho\omega^2}{\kappa G} \frac{d\chi}{dz} \Big|_{z=0,L}. \quad (22)$$

Finally, using Eq. (17), we get

$$\left. \frac{d^3\chi}{dz^3} \right|_{z=0,L} = - \frac{\rho\omega^2}{M_r} \frac{d\chi}{dz} \Big|_{z=0,L}. \quad (23)$$

It is expected that Timoshenko's equation (3) for the variable  $\xi$  with the boundary conditions (20) and (23) should give the same solutions as Timoshenko's equation (3) for the variable  $\psi$  with boundary conditions (16) and (21). Although this seems paradoxical, that this is so as can be easily derived using the relation

$$- \frac{\partial \xi}{\partial z} = \Psi + \gamma_0. \quad (24)$$

More important is that the solutions obtained in this way should be the same as those of Eqs. (1) and (2) with the boundary conditions (13) and (14). One should note that the boundary conditions (20) and (23) depend on the frequency and that they agree with the Bernoulli–Euler boundary conditions [1,4,12] for low frequencies. In what follows, without loss of generality, we will use only the variable  $\chi$ , its equation and the corresponding boundary conditions.

### 2.3. Determination of the wave amplitude

It is possible to obtain the normal-mode frequencies and amplitudes using the same procedure as in Ref. [12], i.e., by substituting directly the solution (12) on the free–free boundary conditions (20) and (23) at  $z=0, L$ . One obtains

$$\begin{pmatrix} (k_+^2 + \frac{\rho\omega^2}{\kappa G}) & (k_+^2 + \frac{\rho\omega^2}{\kappa G}) & (k_-^2 + \frac{\rho\omega^2}{\kappa G}) & (k_-^2 + \frac{\rho\omega^2}{\kappa G}) \\ (k_+^2 + \frac{\rho\omega^2}{\kappa G})e^{k_+L} & (k_+^2 + \frac{\rho\omega^2}{\kappa G})e^{-k_+L} & (k_-^2 + \frac{\rho\omega^2}{\kappa G})e^{k_-L} & (k_-^2 + \frac{\rho\omega^2}{\kappa G})e^{-k_-L} \\ (k_+^3 + \frac{\rho\omega^2 k_+}{M_r}) & -(k_+^3 + \frac{\rho\omega^2 k_+}{M_r}) & (k_-^3 + \frac{\rho\omega^2 k_-}{M_r}) & -(k_-^3 + \frac{\rho\omega^2 k_-}{M_r}) \\ (k_+^3 + \frac{\rho\omega^2 k_+}{M_r})e^{k_+L} & -(k_+^3 + \frac{\rho\omega^2 k_+}{M_r})e^{-k_+L} & (k_-^3 + \frac{\rho\omega^2 k_-}{M_r})e^{k_-L} & -(k_-^3 + \frac{\rho\omega^2 k_-}{M_r})e^{-k_-L} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = 0. \quad (25)$$

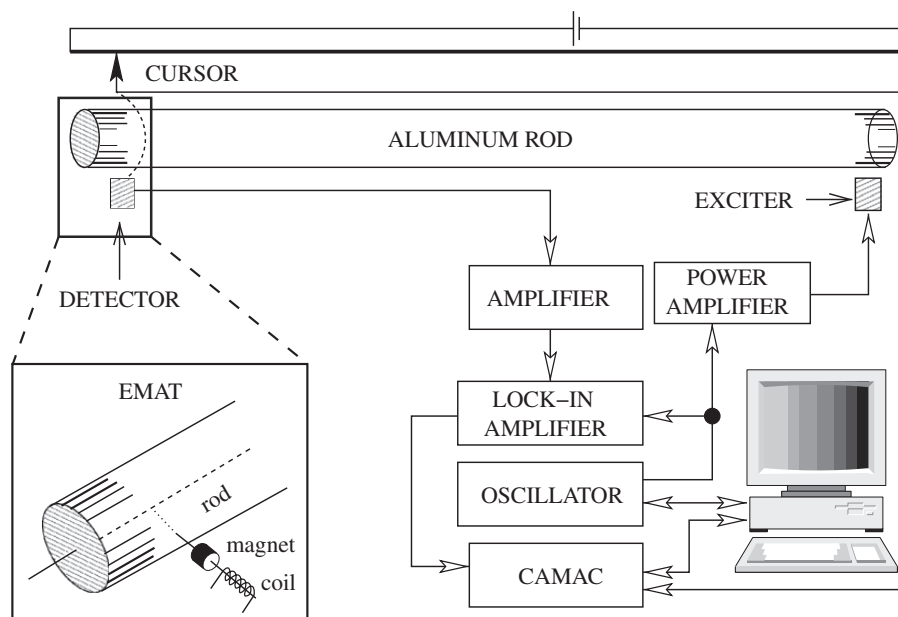
This system of equations has non-trivial solutions when the determinant of the matrix vanishes. One should notice that the determinant is different according to whether  $\omega$  is larger or smaller than  $\omega_c$ .

### 3. Experimental method

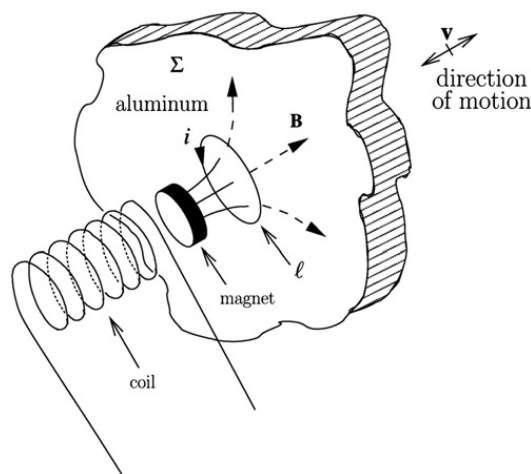
In this section we describe the experimental setup that we have used to obtain the frequencies and amplitudes of oscillation of the flexural modes. With the intention of being able to affirm with all certainty that the measured vibrations correspond to flexural waves, we obtained, both numerically and experimentally, the other two types of vibrations in the rod: torsional and compressional. By comparing the experimental results with the numerical values for these two types of waves, we were able without any doubt to identify the flexural vibrations. In this way, a reliable method has been established.

One of the main problems when measuring elastic vibrations is the lack of selectivity of the detectors for the different types of waves. In order to solve this problem we have used the experimental setup shown in Fig. 2. At the heart of the setup is an electromagnetic acoustic transducer (EMAT) which consists of a permanent magnet and a coil as indicated in Fig. 3. This EMAT has several interesting properties: it can be used both as an exciter as well as a detector; it can be easily constructed and operates from very low frequencies up to hundreds of kHz; and, last but not least for our purposes, it is sensible to the direction of motion of the metallic surface under observation. We shall now discuss this point.

Let us deal first with the detector configuration of Fig. 3 in which the direction of the velocity  $\mathbf{v}$  of the conductor's surface lies along the permanent magnet axis. This magnet produces a divergent magnetic field  $\mathbf{B}$  which induces eddy



**Fig. 2.** Experimental setup used to measure the normal-mode frequencies and wave amplitudes in the aluminum rod. In the left-lower corner the detector is shown for flexural vibrations. The aluminum rod has a length  $L=1.000$  m and radius  $R=0.0127$  m; the elastic constants are  $G=26.92$  GPa,  $E=67.42$  GPa and  $\rho=2699.04$  kg/m<sup>3</sup>.



**Fig. 3.** When the velocity of the surface  $\Sigma$  is parallel to the permanent magnet axis the eddy currents induce an oscillating magnetic field  $\mathbf{B}_i$  (not shown) also parallel to the axis.

currents in the conductor. These currents are excited due to the variation of the magnetic flux in the loop  $\ell$  which oscillates with the material. The magnetic field  $\mathbf{B}_i$  generated by the eddy currents induces an emf in the coil; this voltage provides the signal that detects the oscillation.

To employ the device of Fig. 3 as an exciter, an AC electric current is now sent through the coil. The magnetic field generated by the coil will induce eddy currents on the conductor's surface. According to Lorentz force, due to the presence of the magnetic field  $\mathbf{B}$  there is an applied force on the electrons forming the eddy currents. This force, perpendicular to the surface, is proportional to  $\mathbf{v}_e \times \mathbf{B}_s$  with  $\mathbf{v}_e$  the velocity of the electrons and  $\mathbf{B}_s$  the component of the magnetic field  $\mathbf{B}$  parallel to the surface. Thus, the electrons feel an applied force perpendicular to  $\mathbf{v}_e$ , so the eddy currents give rise to an oscillating force normal to the surface and therefore parallel to the permanent magnet axis.

When the direction of motion is parallel to the surface and perpendicular to the magnet axis, as in the device of Fig. 4 acting as a detector, we consider the loop  $\ell$  on the XY-plane. Since the direction of motion is along the Z-axis, there is a variable magnetic flux across the loop  $\ell$  which induces eddy currents on this loop. The variable magnetic flux is produced by the Z component of  $\mathbf{B}$  which increases in the positive direction of the Z-axis. The eddy current  $i$ , generated by this variable magnetic flux, in turn produces a variable magnetic field  $\mathbf{B}_i$  which induces an emf on the coil; this voltage provides the signal that detects the oscillation.

To convert the device in Fig. 4 into an exciter, an AC current is sent through the coil. The AC magnetic field generated by the coil induces a current on the loop  $\ell$ . Due to the magnetic part of the Lorentz force, the X projection  $\mathbf{B}_x$  of the magnetic field  $\mathbf{B}$ , acting on the electrons produces a force in the Z direction proportional to  $\mathbf{v}_y \times \mathbf{B}_x$ , with  $\mathbf{v}_y$  the Y projection of the electrons velocity. It must be observed that the simultaneous inversions of sign in  $\mathbf{v}_y$  and  $\mathbf{B}_x$  give rise to a force with the same sign along the path  $\ell$ .

A general conclusion can be obtained from the previous discussion: as a detector, the magnetic field  $\mathbf{B}_i$  produced by the motion of the conductor's surface is parallel to the velocity of the latter. A similar conclusion for the device acting as an exciter can be reached: the applied force is parallel to the coil axis of Fig. 4. To verify the above qualitative conclusion for the detector, we have measured the induced voltage in the coil as a function of  $\theta$ , the angle between the direction of motion, in this case along the Z-axis, and the coil axis, as shown in Fig. 5. In Fig. 6 the change of the coil voltage is given as a function of  $\theta$ , that is, the coil axis is rotated with respect to the Y-axis of Fig. 5. As can be seen, this voltage takes a maximum value when  $\theta$  is close to zero, in accordance with our previous conclusion.

We have shown that the EMAT is capable of distinguishing between different modes of oscillation of the conductor. The relative orientations of the coil and permanent magnet detect different oscillations as indicated in Fig. 7. Here configurations  $C_1$  and  $C_2$  detect compressional waves, whereas  $T_1$  and  $T_2$  do so for torsional oscillations. On the other hand,  $F_1$  excites and detects the displacement  $\xi$  of flexural vibrations and the configuration  $F_2$  the angular variable  $\psi$  of

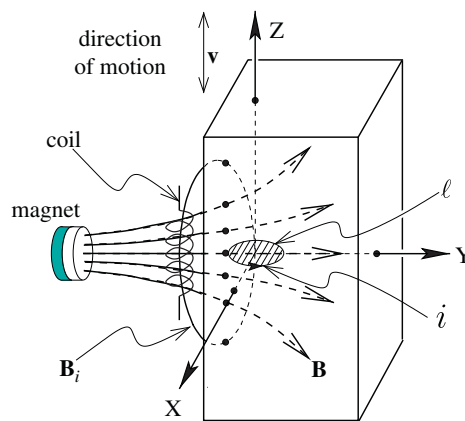


Fig. 4. The direction of motion is perpendicular to the permanent magnet axis. The induced magnetic field is now parallel to the surface velocity.

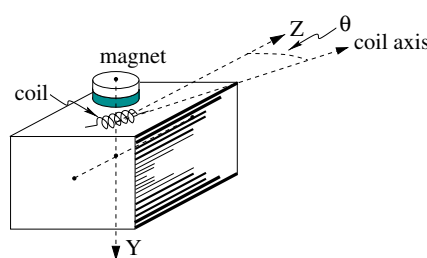
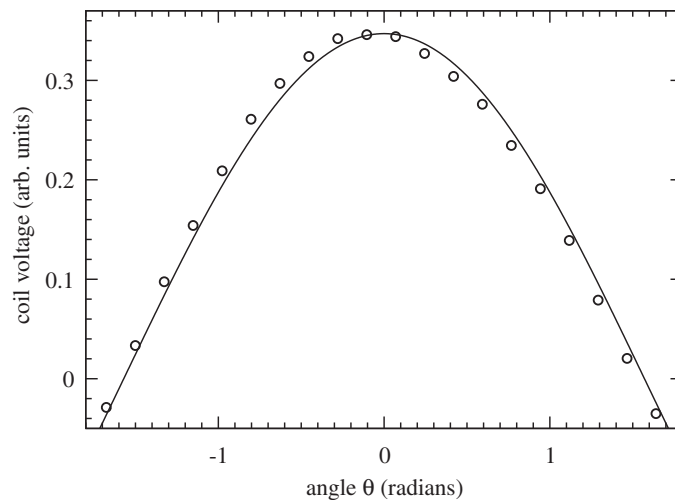
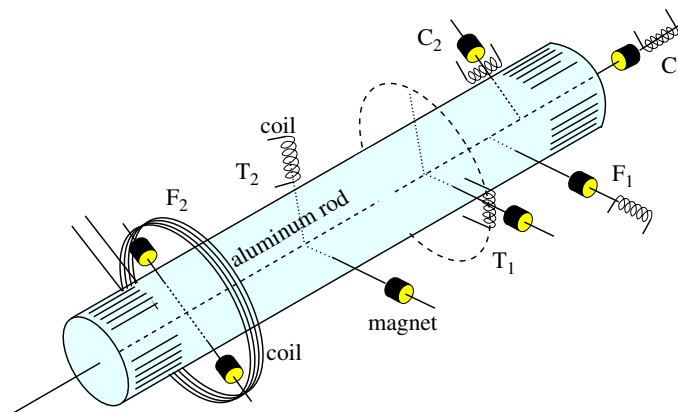


Fig. 5. Experimental setup used to measure the coil voltage as a function of the angle between the coil and the direction of motion. The direction of motion is along the Z-axis.





**Fig. 6.** The circles correspond to experimental values of the coil voltage as a function of the angle  $\theta$  of Fig. 5. The solid line equation is  $0.35 \cos \theta$ .



**Fig. 7.** Different relative positions of the EMAT's coil and magnet excite and detect distinct modes of the rods: compressional vibrations,  $C_1$  and  $C_2$ ; torsional waves  $T_1$  and  $T_2$ ;  $F_1$  and  $F_2$  do so for flexural vibrations.

**Fig. 1.** We conclude this discussion about the EMATs mentioning that they can measure the complete velocity vector on the conductors' surface and that acting as exciters they can produce an arbitrary vector force.

#### 4. Results for frequencies below the critical frequency

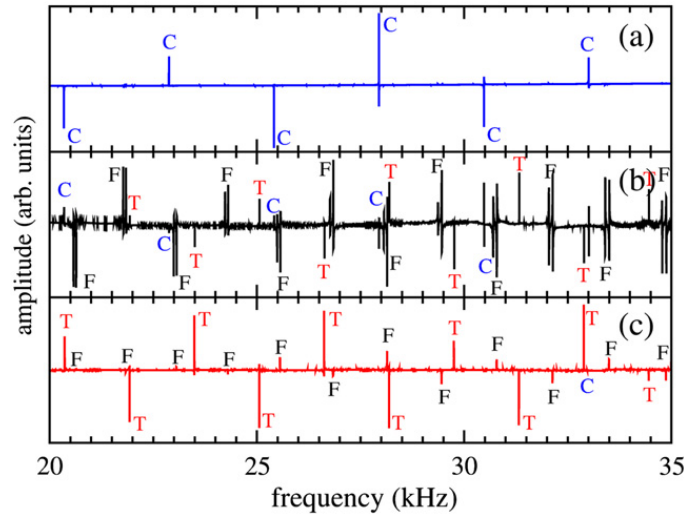
We shall now use the different configurations of Fig. 7 to excite and detect all normal modes of a cylindrical rod and compare the experimental results with the values obtained for free–free boundary conditions with the method described in Section 2 for flexural vibrations and in Ref. [27] for the two other modes. We are mainly interested in the comparison of measured values of the normal-mode frequencies and wave amplitudes with the TBT predictions. In Fig. 8 a typical frequency spectrum is given for the range 20–35 kHz, which is below the critical frequency  $f_c \approx 150$  kHz. All the peaks have been identified and associated, one to one, with either compressional (C), torsional (T), or flexural (F) normal modes. One can observe that some lines of spectra (a) and (c) are also seen, with a significant intensity, in spectrum (b). This is due to the fact that the transversal component of the compressional wave is excited and registered by the flexural configuration  $F_1$ . Something similar happens with the torsional states marked T in the figure.

To further identify the flexural normal modes we make two remarks. First, all flexural lines in Fig. 8(b) appear as doublets [4]. This is due to the fact that the cross-section of the rod is not a perfect circle, so two principal values of the moment of inertia are present in the eigenvalue equations of Section 2. Therefore, two orthogonal polarized eigenmodes, with slightly different frequencies, are excited. The differences in frequency for the doublet lines observed in the experiment are consistent with the irregularities that we measured in the rod radius. This is the main reason for using a cylindrical rod in this part of our work.

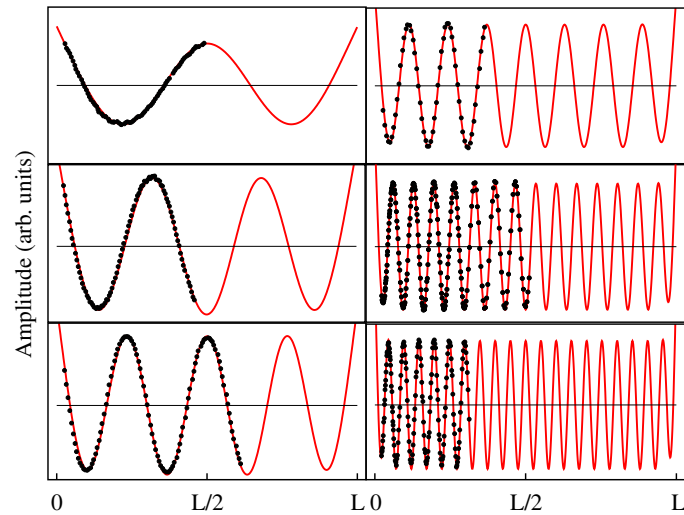
Second, to distinguish flexural from compressional modes, we note that the wave amplitude for the latter is constant along the free cross-section, whereas for flexural vibrations the amplitude changes sign along a diameter. We have verified experimentally that this is indeed the case.

Once we were certain that a peak corresponds indeed to a flexural normal mode, we measured the corresponding wave amplitude sliding the detector along the rod and registering its position, using the experimental setup of Fig. 2. Six





**Fig. 8.** In (a) and (c) the picket-fence experimental spectra corresponding, respectively, to compressional (C) and torsional (T) waves are given. In (b) the doublets related to flexural (F) vibrations are shown. Note that in spectrum (b) states C and T also appear.



**Fig. 9.** Wave amplitudes for  $N = 4, 6, 8, 16, 30$  and  $40$  nodes for the rod used in Fig. 8 with a length  $L = 1.000$  m. The experimental wave amplitudes are given by the dots. The continuous lines correspond to the theoretical prediction equation (12).

examples of normal modes are shown in Fig. 9; the dots correspond to experimental measurements and the continuous lines are obtained from TBT.

To test the dispersion relation we have to obtain  $k_+$  and  $k_- = -ik_+$  as a function of the frequency  $f$  (see Eq. (11)). The values of the wavenumbers  $k_{+,n}$  and  $k_n$  and frequencies  $f_n$  can then be obtained from the  $n$ -th normal-mode wave amplitude and frequency, respectively. Disregarding the growing solution, the experimental wave amplitudes below  $f_c$  can be fitted by

$$\chi(z) = A_1 \cos(k_- z + \phi) + A_2 e^{-k_+ z}, \quad (26)$$

where  $A_1$  and  $A_2$  are constants to be determined. Notice that only a portion of the wave amplitude is used to get both  $k_+$  and  $k_-$ ; this is due to the following fact: when both EMAT's are close to each other, the magnetic field of the EMAT used as exciter saturates the coil of the EMAT used as detector yielding an interference offset that depends on the distance between both EMAT's. The nonlinear fitting (26) was used to obtain the dispersion relation and it was tested with numerical wave amplitudes of both Bernoulli–Euler and Timoshenko equations. In the case of the wave amplitude with 10 nodes the nonlinear fitting gives the theoretical wavenumbers with five digits (of precision). The measurements corresponding to the first 2 cm were erased to avoid artifacts due to the changes of the eddy currents near the free-ends of the rod. In Fig. 10 we show that the wavenumbers given by Eq. (11) agree with those obtained with the nonlinear fitting of the experimental wave amplitudes, the error ranging from 0.001 percent to 1.57 percent. The maximum error is obtained for the lowest modes since the growing exponential is then nonnegligible in the experimental wave amplitude and is not included in the nonlinear fitting of Eq. (26). Nevertheless, an excellent agreement is obtained for  $f < f_c$  with the prediction of TBT.

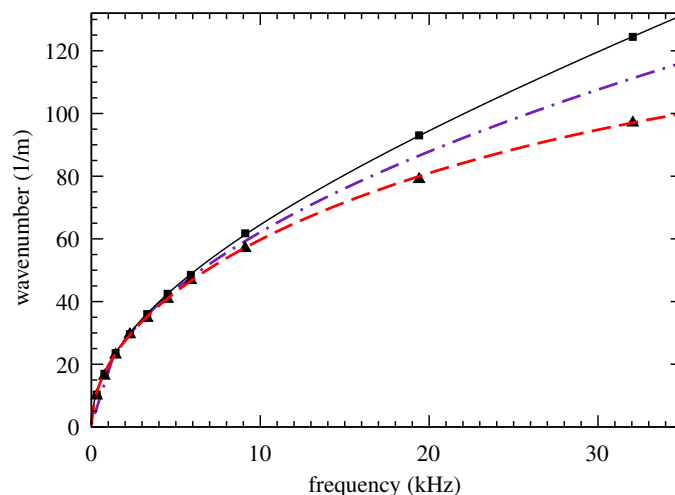
## 5. Results above the critical frequency

We dedicate this section to compare TBT and the finite element method (FEM) calculations with experimental results for frequencies beyond the critical frequency for a rectangular beam which shows a simpler spectrum than what is obtained for a cylindrical rod. We show a very good coincidence in these three cases, so one can conclude that the second spectrum of Timoshenko indeed exists, at least for the free-free boundary conditions case.

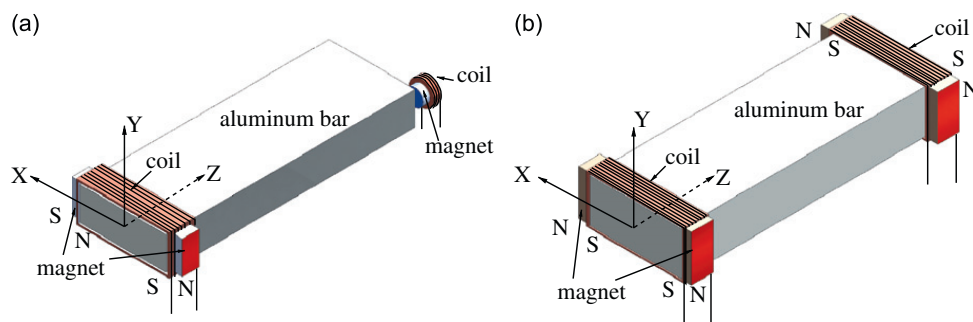
In Fig. 11(a) the configuration that we have used to excite and detect the bending modes of the beam is shown: First, the EMAT on the right-hand side used as exciter, applies a force perpendicular to the coil axis. Since this exciter is located nonsymmetrically with respect to the face of the beam (that is, close to a corner), it excites flexural waves on both the soft and hard sides of the beam, as well as compressional waves. Second, the EMAT on the left-hand side of the beam, due to the configuration of the magnets, detects the flexural modes on the hard side of the beam. Notice that in spite of the fact that the detector configuration is set to detect flexural waves the compressional modes, easier to be excited, are also detected.

To determine which resonant peaks of the spectrum of Fig. 12(a) correspond to compressional resonances and which to flexural ones, the compressional resonances were measured using the configuration shown in Fig. 11(b). The corresponding compressional spectrum is shown in Fig. 12(b). The resonant bending spectrum of the stiff plane of the beam is obtained subtracting the compressional peaks given in the lower part of Fig. 12 from the spectrum given in the upper part of the same figure.

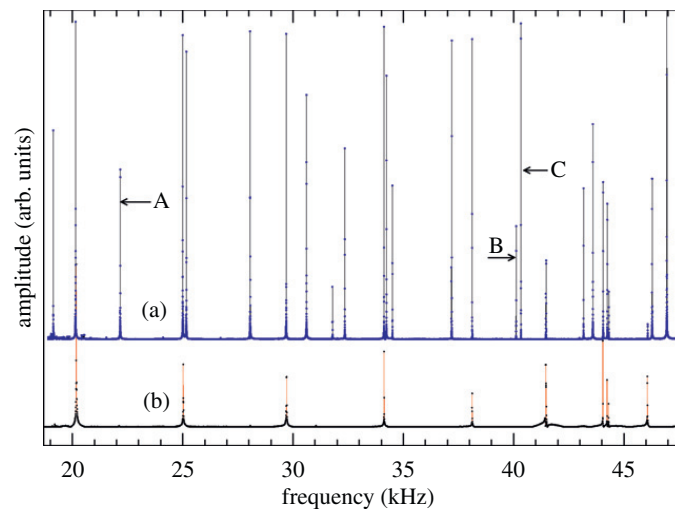
To compare with the experimental and TBT predictions, a numerical analysis was performed using the FEM platform Salome-Meca [28]. The beam was modeled using 3D Tetrahedron (Netgen). The simulations were performed with the physical parameters given in Fig. 11. The TBT as well as the numerical analysis use, as input, several physical parameters. They were obtained from the experiment as follows. On the one hand, the shear and Young's moduli were obtained from the torsional and compressional velocity measuring the lowest torsional and compressional modes of the beam, respectively. The value of the Timoshenko shear coefficient, on the other hand, was obtained from the best fit to the critical frequency.



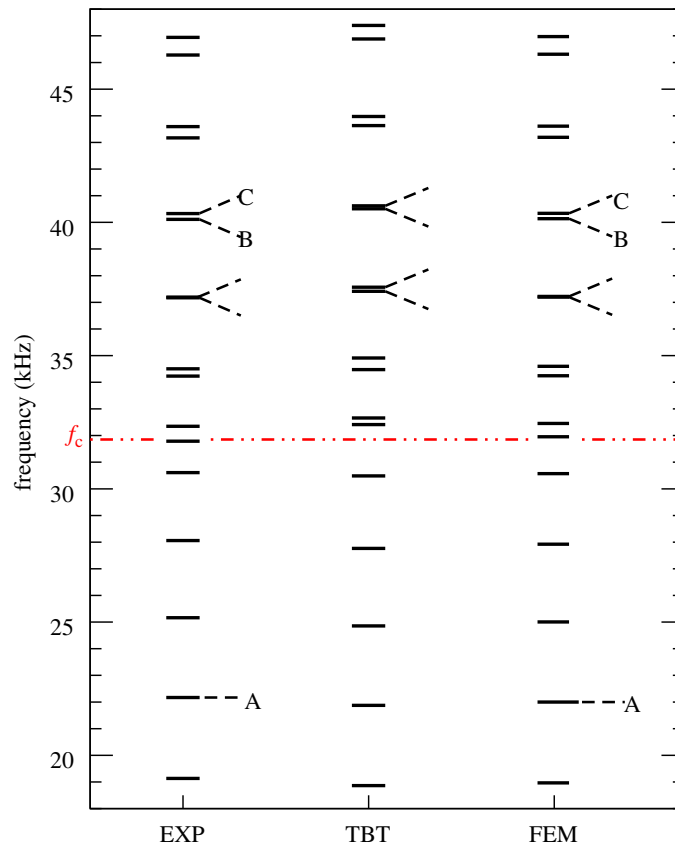
**Fig. 10.** Experimental values obtained for  $k_+$  (squares) and  $k_-$  (triangles) compared with the TBT values, respectively, the solid and dashed lines. The intermediate dot-dashed line corresponds to the Bernoulli-Euler theory.



**Fig. 11.** Experimental setup used to excite and detect different elastic waves. The configuration shown in (a) is used to measure flexural waves on the stiff plane of the beam; the EMAT at the left-hand side of the beam corresponds to the detector while the EMAT at the opposite side of the beam generates bending waves in both the flexible and stiff planes as well as compressional modes. The direction of the force applied by the exciter to the beam is along the Z-axis. To discriminate the compressional modes from the spectrum, they were measured using the configuration shown in (b). The aluminum beam of rectangular cross-section has dimension:  $L=0.5$  m, height  $a=0.0252$  m and width  $b=0.0504$  m and the same elastic constants as in Fig. 2. The same parameters were used in the FEM simulations.



**Fig. 12.** Resonant spectrum measured for the beam of Fig. 11. The upper spectrum was measured with the setup of Fig. 11(a) while the lower one was measured with the setup of Fig. 11(b). The stiff bending resonances (such as A, B and C) are those of the upper spectrum that do not appear in the compressional (lower) spectrum.



**Fig. 13.** Flexural stiff spectrum (left column) measured with the EMATs for the beam of Fig. 11. The central column corresponds to the TBT prediction while the right column was obtained using the finite element method platform Salome-Meca. We used  $\kappa = 0.85$  for Timoshenko's shear coefficient. Above the critical frequency all spectra show doublets with levels very close to each other in some cases (made apparent with dotted lines). The face deformation of modes A, B and C are shown in Fig. 15.

In Fig. 13 we show the comparison between the spectrum of the measured bending modes with the spectrum calculated numerically using Salome-Meca as well as with the prediction of TBT. The cut-off frequencies are  $f_{c,F} = 63.70$  kHz and  $f_{c,S} = 31.850$  and correspond to the vibrations in the more flexible (F) and in the stiff (S) planes, respectively. They were calculated using Eq. (9). Below the critical frequency the FEM calculations and TBT predictions agree with the experiment with errors less than  $\approx 2$  percent. Above the critical frequency all spectra show doublets but

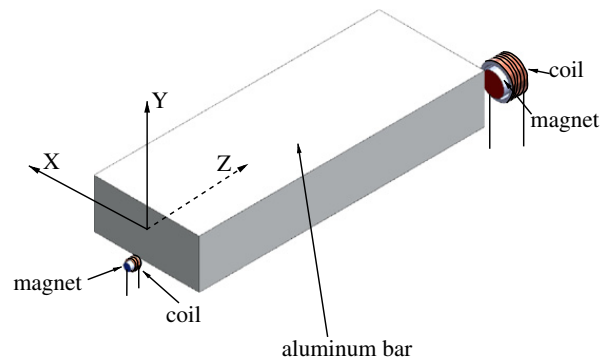
the predictions of TBT show errors again less than  $\approx 2$  percent. This figure demonstrates that the TBT correctly predicts the spectrum above the critical frequency at least for free-free boundary conditions. One can also observe in Fig. 13 that the FEM calculations agree with the experimental results. Our experimental and numerical results for the normal stiff mode frequencies of the beam are summarized in Table 1. One can see that the error between experimental results and TBT or FEM results are about 2 percent for the fundamental mode. To get a better approximation for the lowest flexural mode one has to use the value of the compressional velocity ( $=5090.23$  m/s) for very low frequencies; this yields an error of 0.25 percent for the lowest flexural mode but an error of about 2 percent for higher modes.

We have also measured with the experimental setup of Fig. 14 the wave amplitude of the bending modes along the X-axis of the face  $Z=0$  shown in the same figure. In the lower panels of Fig. 15 we give the wave amplitudes for the three states labeled A, B and C in Fig. 13. State A corresponds to a mode with frequency below the critical one, while wave amplitudes B and C belong to a doublet above  $f_c$ . In the upper part of Fig. 15, the cross-section wave amplitude calculated with Salome-Meca for these three resonances are shown and, as before, the one labeled A corresponds to a frequency below the critical frequency and those with labels B and C to a doublet above the critical frequency.

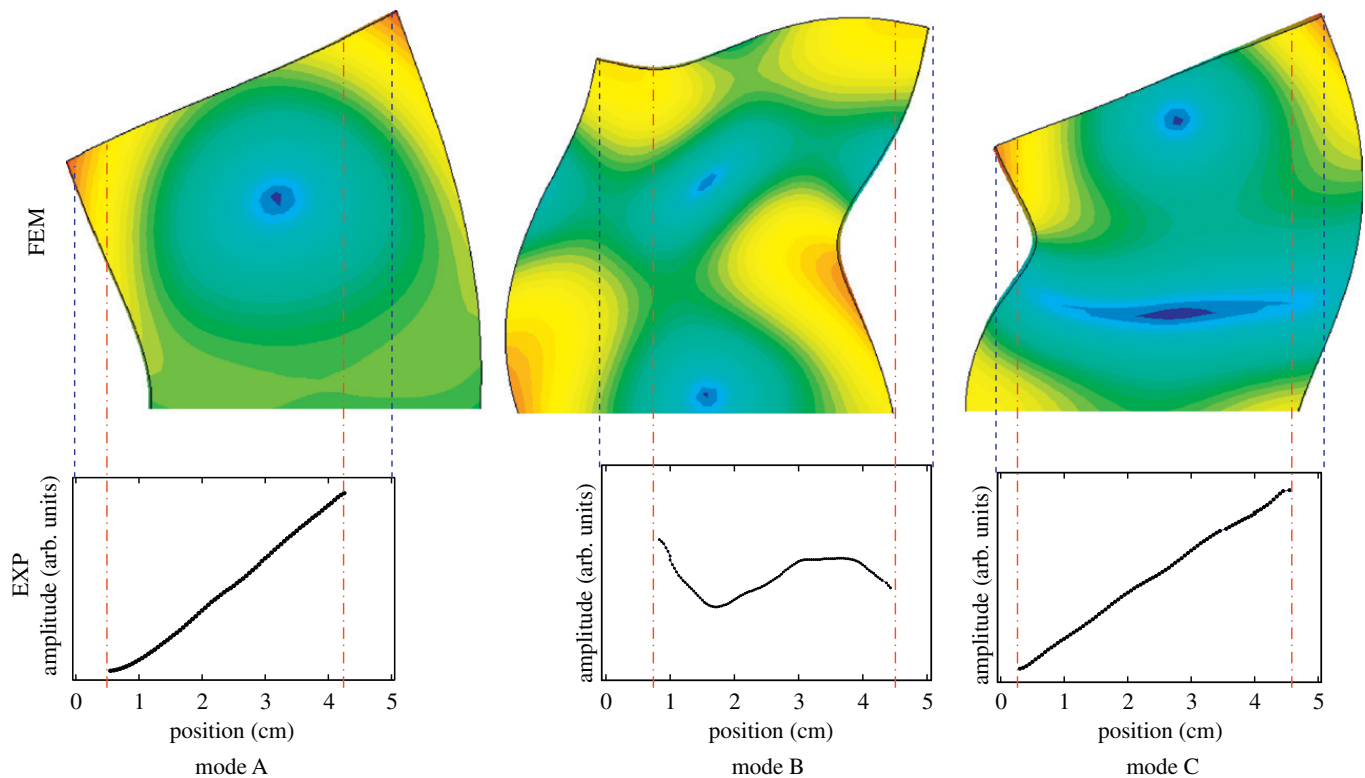
In Timoshenko beam theory the surface of the face is assumed to be plane. This is true for all amplitudes we have measured below the critical frequency. The same result appears for all wave amplitudes calculated with Salome-Meca below  $f_c$ . Above the critical frequency the results are completely different. One amplitude of each doublet is plane but the other is almost sinusoidal. This result was checked for several doublets and the results of the FEM agree with the experiment. Nevertheless, the frequency values from TBT agree with experiment.

**Table 1**  
Normal mode frequencies for the beam of Fig. 11 with free ends.

| Mode   | Experiment (kHz) | TBT (kHz)        | FEM (kHz)       |
|--------|------------------|------------------|-----------------|
| 1      | 1.0211           | 1.00062 (+2.01%) | 1.0008 (+1.99%) |
| 2      | 2.6594           | 2.60559 (+2.02%) | 2.6076 (+1.95%) |
| 3      | 4.8462           | 4.75908 (+1.80%) | 4.7664 (+1.65%) |
| 4      | 7.3878           | 7.27210 (+1.57%) | 7.2893 (+1.33%) |
| 5      | 10.163           | 10.0153 (+1.45%) | 10.047 (+1.14%) |
| 6      | 13.082           | 12.8989 (+1.40%) | 12.950 (+1.01%) |
| 7      | 16.081           | 15.8629 (+1.36%) | 15.938 (+0.89%) |
| 8      | 19.1335          | 18.8651 (+1.40%) | 18.966 (+0.88%) |
| 9 (A)  | 22.1699          | 21.8731 (+1.34%) | 22.000 (+0.77%) |
| 10     | 25.1638          | 24.8548 (+1.23%) | 25.004 (+0.64%) |
| 11     | 28.0593          | 27.7656 (+1.05%) | 27.923 (+0.49%) |
| 12     | 30.6109          | 30.4859 (+0.41%) | 30.570 (+0.13%) |
| $f_c$  |                  | 31.850           |                 |
| 13     | 31.7895          | 32.4096 (−1.94%) | 31.954 (−0.51%) |
| 14     | 32.3453          | 32.6607 (−0.97%) | 32.453 (−0.32%) |
| 15     | 34.2301          | 34.4741 (−0.70%) | 34.245 (−0.03%) |
| 16     | 34.5053          | 34.9106 (−1.16%) | 34.597 (−0.26%) |
| 17     | 37.1713          | 37.4119 (−0.64%) | 37.196 (−0.06%) |
| 18     | 37.1940          | 37.5655 (−0.99%) | 37.222 (−0.07%) |
| 19 (B) | 40.1110          | 40.5102 (−0.99%) | 40.136 (−0.05%) |
| 20 (C) | 40.3298          | 40.6225 (−0.72%) | 40.339 (−0.01%) |
| 21     | 43.1702          | 43.6314 (−1.06%) | 43.191 (−0.04%) |
| 22     | 43.5919          | 43.9734 (−0.87%) | 43.611 (−0.03%) |
| 23     | 46.2782          | 46.8812 (−1.29%) | 46.310 (−0.06%) |
| 24     | 46.9439          | 47.3915 (−0.94%) | 46.973 (−0.05%) |



**Fig. 14.** Experimental setup used to measure the wave amplitude on the face of the beam. The detector (left-hand side) was moved along the X-axis. The position of the exciter (right-hand side) is fixed.



**Fig. 15.** Wave amplitudes of a mode A below the critical frequency and of a doublet above  $f_c$ , modes B and C; the frequencies are given in Table 1. The measurements (lower panels) were done moving the detector along the X-axis using the experimental setup of Fig. 14. The upper panels are the results obtained with Salome-Meca using the same parameters of Fig. 11. Note that the deformations calculated with FEM agree with the experimental measurements shown in the lower panels. The measurement at the borders of the beam are affected by an edge effect of the order of the size of the detector, which is approximately 0.25 cm, and therefore they are not shown.

## 6. Conclusions

In this paper we have studied, from the theoretical and experimental points of view, the flexural normal-mode frequencies and wave amplitudes of cylindrical rods and rectangular beams for free–free boundary conditions. We consider two frequency regimes: frequencies less than the critical one  $f_c$  and for  $f$  beyond  $f_c$ . For the first regime, TBT is much more than a beautiful engineering theory, it is also a very accurate one. This fact allowed us to test our experimental procedure. We obtain an excellent agreement between TBT and the measurements for frequencies, wave amplitudes and the dispersion relation. These, as well as the results for the second regime, were corroborated with 3-D FEM calculations using the Salome-Meca platform. Therefore, the results presented in Section 4 serve to validate our experimental procedure. We have also performed experiments for the second regime  $f > f_c$ , experiments which have a wider perspective than what has been achieved up to now. For a rectangular beam the flexural modes, and in this frequency regime, appear always as doublets.<sup>1</sup> Since we can also measure the deformation of the free face of the beam, we have proven that one of the states of the doublet shows a flat surface while the other state has a sinusoidal shape. This is consistent with the exact Rayleigh–Lamb solution for the transverse field shape of the second-branch mode. For  $f < f_c$ , on the other hand, all normal modes show a flat free face, an assumption made when deriving TBT equations. These facts have also been observed in the experiments. In any case, we can assure that the second spectrum of Timoshenko exists, at least for free–free boundary conditions. The TBT predictions are also very accurate beyond about 50 percent of the critical frequency; this is consistent with Ref. [25] in which the TBT dispersion relation, slightly modified by the inertia correction factor as well as the shear correction factor, agrees with the exact Rayleigh–Lamb prediction up to frequencies of the order of twice the critical frequency. Although the TBT assumes a flat surface, even for frequencies beyond the critical one, it yields very precise results (with an error less than 2 percent) for the normal-mode frequencies. Something similar happens with the Rayleigh–Ritz method that produces extremely accurate results for the normal-mode frequencies with very poor trial wave functions.

Although the results presented here are conclusive about the existence of the second TBT spectrum, there are two open questions left. The first one, related to the normal-mode frequencies, is to find out the upper limit for which the Timoshenko beam theory is valid. That work requires to study much higher frequencies to distinguish the TBT resonances

<sup>1</sup> We should note that the origin of these doublets is different from what was obtained in Section 4 for the cylindrical rod. We have also measured the spectrum for  $f > f_c$  for a rod which is almost perfectly circular; the spectrum consists of quadruplets.

from the three-dimensional spectrum. The second question is related to the dispersion relation. To obtain it the wave amplitudes above the critical frequency must be measured.

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- [28] The program Salome-Meca is a GNU integration platform based on FEM with GNU Generic Public License (GPL). It was developed by Électricité de France (EDF) and tested in many instances. <<http://www.code-aster.org/>>.